AIR POLLUTION AND CPCB STANDARDS

AMBIENT-EMISSION FIREWALL → CONCENTRATION-TO-EXPOSURE CHAIN → POLLUTANT FORMATION MAP → PM CLASSIFICATION GATE → SMOG AND ACID-DEPOSITION MATRIX → NAAQS DATA GATE

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

◆ PRIMARY CORE FLOW ◆ SUBORDINATE EXTRA ◆ APPROVAL / REVIEW

CORE BEFORE EXTRA • prior artifacts unchanged • exact same master drives poster and tiles

Approval: FALSE • Pending user review • active generation g2 • explicit approval of this exact generation is required • all prior artifacts unchanged

00

STAGE 00 AMBIENT-EMISSION FIREWALL

AMBIENT-EMISSION FIREWALL · AMBIENT STANDARD · CONCENTRATION IN OUTDOOR RECEIVING AIR · SOURCE STANDARD · RELEASE FROM A NAMED STACK, VEHICLE OR PROCESS · CONSENT CONDITION · SOURCE-SPECIFIC LEGAL CONTROL · PUBLIC COMMUNICATION INDEX

ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
AMBIENT STANDARD → concentration in outdoor receiving air	CONSENT CONDITION → source-specific legal control	RULE → never substitute one instrument for another
SOURCE STANDARD → release from a named stack, vehicle or process	AQI → public communication index	AMBIENT STANDARD → concentration in outdoor receiving air

ECOLOGICAL UNIT / PARAMETER
- AMBIENT STANDARD → concentration in outdoor receiving air
- SOURCE STANDARD → release from a named stack, vehicle or process

DECISIVE DISTINCTION
- CONSENT CONDITION → source-specific legal control
- AQI → public communication index

SCALE / STATUS / EXCEPTION
- RULE → never substitute one instrument for another
- AMBIENT STANDARD → concentration in outdoor receiving air

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → never substitute one instrument for another.

01

STAGE 01 CONCENTRATION-TO-EXPOSURE CHAIN

CONCENTRATION-TO-EXPOSURE CHAIN · POLLUTANT CONCENTRATION AT PLACE AND TIME · AVERAGING PERIOD · FIXES THE MEASUREMENT WINDOW · PERSON LOCATION · DETERMINES CONTACT OPPORTUNITY · DURATION AND ROUTE · SHAPE EXPOSURE

MONITOR → pollutant concentration at place and time → AVERAGING PERIOD → fixes the measurement window → PERSON LOCATION → determines contact opportunity → DURATION AND ROUTE → shape exposure → HEALTH CLAIM → needs separate epidemiological evidence

SYSTEM INPUT / STARTING CONDITION
- MONITOR → pollutant concentration at place and time
- AVERAGING PERIOD → fixes the measurement window

ECOLOGICAL PROCESS / TRANSFER
- PERSON LOCATION → determines contact opportunity
- DURATION AND ROUTE → shape exposure

OUTCOME / FEEDBACK / LIMIT
- HEALTH CLAIM → needs separate epidemiological evidence
- MONITOR → pollutant concentration at place and time

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MONITOR → pollutant concentration at place and time.

02

STAGE 02 POLLUTANT FORMATION MAP

POLLUTANT FORMATION MAP · PRIMARY EMISSION · RELEASED DIRECTLY · ENTERS ATMOSPHERIC REACTION · SUNLIGHT OR OXIDATION · TRANSFORMATION PATHWAY · SECONDARY POLLUTANT · FORMED IN AIR

PRIMARY EMISSION → released directly

PRECURSOR → enters atmospheric reaction → SUNLIGHT OR OXIDATION → transformation pathway → SECONDARY POLLUTANT → formed in air

SYSTEM INPUT / STARTING CONDITION
- PRIMARY EMISSION → released directly
- PRECURSOR → enters atmospheric reaction

ECOLOGICAL PROCESS / TRANSFER
- SUNLIGHT OR OXIDATION → transformation pathway
- SECONDARY POLLUTANT → formed in air

OUTCOME / FEEDBACK / LIMIT
- CONTROL → address direct emissions and precursors
- PRIMARY EMISSION → released directly

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SECONDARY POLLUTANT → formed in air; CONTROL → address direct emissions and precursors.

03

STAGE 03 PM CLASSIFICATION GATE

PM CLASSIFICATION GATE · PM2.5 OR PM10 · AERODYNAMIC SIZE FRACTION · AFFECTS TRANSPORT AND DEPOSITION · REQUIRES CHEMICAL EVIDENCE · REQUIRES APPORTIONMENT EVIDENCE · SIZE LABEL ALONE PROVES NEITHER COMPOSITION NOR SOURCE

PM2.5 OR PM10 → aerodynamic size fraction

SIZE → affects transport and deposition → COMPOSITION → requires chemical evidence → SOURCE → requires apportionment evidence

SYSTEM / LEVEL / DEFINITION
- PM2.5 OR PM10 → aerodynamic size fraction
- SIZE → affects transport and deposition

STRUCTURE-FUNCTION MECHANISM
- COMPOSITION → requires chemical evidence
- SOURCE → requires apportionment evidence

SCALE / EVIDENCE LIMIT
- TRAP → size label alone proves neither composition nor source
- PM2.5 OR PM10 → aerodynamic size fraction

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → size label alone proves neither composition nor source.

04

STAGE 04 SMOG AND ACID-DEPOSITION MATRIX

SMOG AND ACID-DEPOSITION MATRIX · PRECURSOR EMISSIONS PLUS SUNLIGHT-DRIVEN CHEMISTRY · SURFACE OZONE · SECONDARY OXIDANT · ACID DEPOSITION · SULPHUR AND NITROGEN PRECURSOR PATHWAY · WET OR DRY · DEPOSITION ROUTES

ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
SMOG → precursor emissions plus sunlight-driven chemistry	ACID DEPOSITION → sulphur and nitrogen precursor pathway	RULE → precursor is not the final atmospheric product
SURFACE OZONE → secondary oxidant	WET OR DRY → deposition routes	SMOG → precursor emissions plus sunlight-driven chemistry

ECOLOGICAL UNIT / PARAMETER
- SMOG → precursor emissions plus sunlight-driven chemistry
- SURFACE OZONE → secondary oxidant

DECISIVE DISTINCTION
- ACID DEPOSITION → sulphur and nitrogen precursor pathway
- WET OR DRY → deposition routes

SCALE / STATUS / EXCEPTION
- RULE → precursor is not the final atmospheric product
- SMOG → precursor emissions plus sunlight-driven chemistry

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → precursor is not the final atmospheric product.

05

STAGE 05 NAAQS DATA GATE

NAAQS DATA GATE · IDENTIFY EXACT SUBSTANCE OR FRACTION · PRESERVE THE NOTIFIED UNIT · AVERAGING PERIOD · ANNUAL, DAILY OR SHORTER ONLY IF SOURCED · AREA AND COMPLIANCE RULE · READ NOTIFICATION · NO TABLE

POLLUTANT → identify exact substance or fraction

UNIT → preserve the notified unit → AVERAGING PERIOD → annual, daily or shorter only if sourced → AREA AND COMPLIANCE RULE → read notification

SYSTEM / LEVEL / DEFINITION
- POLLUTANT → identify exact substance or fraction
- UNIT → preserve the notified unit

STRUCTURE-FUNCTION MECHANISM
- AVERAGING PERIOD → annual, daily or shorter only if sourced
- AREA AND COMPLIANCE RULE → read notification

SCALE / EVIDENCE LIMIT
- NO TABLE → no invented limit
- POLLUTANT → identify exact substance or fraction

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: AVERAGING PERIOD → annual, daily or shorter only if sourced.

06

STAGE 06 AQI STATUTORY BOUNDARY

AQI STATUTORY BOUNDARY · MONITORED POLLUTANT CONCENTRATIONS · PUBLIC-FACING CATEGORY · NOTIFIED AMBIENT STANDARD · SOURCE LIMIT · SEPARATE EMISSION CONTROL · AQI CATEGORY IS NOT EITHER STANDARD TABLE

ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
INPUT → monitored pollutant concentrations	NAAQS → notified ambient standard	VERDICT → AQI category is not either standard table
OUTPUT → public-facing category	SOURCE LIMIT → separate emission control	INPUT → monitored pollutant concentrations

ECOLOGICAL UNIT / PARAMETER
- INPUT → monitored pollutant concentrations
- OUTPUT → public-facing category

DECISIVE DISTINCTION
- NAAQS → notified ambient standard
- SOURCE LIMIT → separate emission control

SCALE / STATUS / EXCEPTION
- VERDICT → AQI category is not either standard table
- INPUT → monitored pollutant concentrations

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERDICT → AQI category is not either standard table.

07

STAGE 07 INSTITUTION AND CLEARANCE LADDER

INSTITUTION AND CLEARANCE LADDER · NATIONAL COORDINATION AND FRAMEWORK · SPCB OR PCC · CONSENT, MONITORING AND ENFORCEMENT · CTE OR CTO · SOURCE-OPERATION PERMISSION LAYER · ENVIRONMENTAL CLEARANCE · PRIOR PROJECT-APPRAISAL LAYER

CPCB → national coordination and framework

SPCB OR PCC → consent, monitoring and enforcement → CTE OR CTO → source-operation permission layer → ENVIRONMENTAL CLEARANCE → prior project-appraisal layer

SYSTEM / LEVEL / DEFINITION
- CPCB → national coordination and framework
- SPCB OR PCC → consent, monitoring and enforcement

STRUCTURE-FUNCTION MECHANISM
- CTE OR CTO → source-operation permission layer
- ENVIRONMENTAL CLEARANCE → prior project-appraisal layer

SCALE / EVIDENCE LIMIT
- RULE → approvals remain legally distinct
- CPCB → national coordination and framework

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SPCB OR PCC → consent, monitoring and enforcement.

08

STAGE 08 MONITORING EVIDENCE LADDER

MONITORING EVIDENCE LADDER · ONE LOCATION · SPATIAL COVERAGE · DATA COMPLETENESS · USABLE TIME RECORD · DISPERSION AND ACCUMULATION CONTEXT · SCALE ONLY AS FAR AS EVIDENCE SUPPORTS

STATION → one location

NETWORK → spatial coverage → DATA COMPLETENESS → usable time record → METEOROLOGY → dispersion and accumulation context

SYSTEM / LEVEL / DEFINITION
- STATION → one location
- NETWORK → spatial coverage

STRUCTURE-FUNCTION MECHANISM
- DATA COMPLETENESS → usable time record
- METEOROLOGY → dispersion and accumulation context

SCALE / EVIDENCE LIMIT
- INFERENCE → scale only as far as evidence supports
- STATION → one location

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: INFERENCE → scale only as far as evidence supports.

09

STAGE 09 SOURCE-APPORTIONMENT CUBE

SOURCE-APPORTIONMENT CUBE · SOURCE MIX CHANGES BY LOCATION · ACTIVITY AND METEOROLOGY CHANGE · PM, OZONE AND GASES HAVE DIFFERENT PATHWAYS · MODEL AND SAMPLING ASSUMPTIONS MATTER · NO UNIVERSAL SHARE · DATE AND SCOPE EVERY ESTIMATE

ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
CITY → source mix changes by location	POLLUTANT → PM, ozone and gases have different pathways	NO UNIVERSAL SHARE → date and scope every estimate
SEASON → activity and meteorology change	METHOD → model and sampling assumptions matter	CITY → source mix changes by location

ECOLOGICAL UNIT / PARAMETER
- CITY → source mix changes by location
- SEASON → activity and meteorology change

DECISIVE DISTINCTION
- POLLUTANT → PM, ozone and gases have different pathways
- METHOD → model and sampling assumptions matter

SCALE / STATUS / EXCEPTION
- NO UNIVERSAL SHARE → date and scope every estimate
- CITY → source mix changes by location

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NO UNIVERSAL SHARE → date and scope every estimate.

10

STAGE 10 PROGRAMME-RESPONSE ARCHITECTURE

PROGRAMME-RESPONSE ARCHITECTURE · LONG-RUN STRUCTURAL PLANNING · STANDARDS-BASED CLASSIFICATION OVER A PERIOD · EPISODIC GRADED RESPONSE · STATUTORY DELHI-NCR REGIONAL COORDINATION · NO SUBSTITUTION · EACH SERVES A DIFFERENT FUNCTION

ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
NCAP → long-run structural planning	GRAP → episodic graded response	NO SUBSTITUTION → each serves a different function
NON-ATTAINMENT → standards-based classification over a period	CAQM → statutory Delhi-NCR regional coordination	NCAP → long-run structural planning

ECOLOGICAL UNIT / PARAMETER
- NCAP → long-run structural planning
- NON-ATTAINMENT → standards-based classification over a period

DECISIVE DISTINCTION
- GRAP → episodic graded response
- CAQM → statutory Delhi-NCR regional coordination

SCALE / STATUS / EXCEPTION
- NO SUBSTITUTION → each serves a different function
- NCAP → long-run structural planning

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NO SUBSTITUTION → each serves a different function.

11

STAGE 11 EVIDENCE AND ANSWER SPINE

EVIDENCE AND ANSWER SPINE · AMBIENT, EMISSION, CONCENTRATION AND EXPOSURE · SOURCE, PRECURSOR, ATMOSPHERIC FORMATION AND RECEPTOR · NAAQS, AQI, CONSENT, NCAP, GRAP AND CAQM · STATION, AVERAGING PERIOD AND APPORTIONMENT BOUNDARY · TARGET, ATTAINMENT, CURRENT DATA AND PYQ KEY STAY DISTINCT

DEFINE → ambient, emission, concentration and exposure → TRACE → source, precursor, atmospheric formation and receptor → GOVERN → NAAQS, AQI, consent, NCAP, GRAP and CAQM → MEASURE → station, averaging period and apportionment boundary → AUDIT → target, attainment, current data and PYQ key stay distinct

SYSTEM / LEVEL / DEFINITION
- DEFINE → ambient, emission, concentration and exposure
- TRACE → source, precursor, atmospheric formation and receptor

STRUCTURE-FUNCTION MECHANISM
- GOVERN → NAAQS, AQI, consent, NCAP, GRAP and CAQM
- MEASURE → station, averaging period and apportionment boundary

SCALE / EVIDENCE LIMIT
- AUDIT → target, attainment, current data and PYQ key stay distinct
- DEFINE → ambient, emission, concentration and exposure

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: AUDIT → target, attainment, current data and PYQ key stay distinct.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · EFFECTIVE AIR-POLLUTION POLICY DEPENDS ON ACCURATE SOURCE APPORTIONMENT (SCIENTIFICALLY · COMMON ANALYTICAL ERROR TO AVOID · TREATING CROP-RESIDUE BURNING AS THE DOMINANT · NCAP SET CITY-SPECIFIC PARTICULATE-MATTER (PM2.5/PM10) REDUCTION TARGETS FOR IDENTIFIED · IMPLEMENTATION CRITIQUE · PROGRESS TRACKING UNDER NCAP HAS FACED CHALLENGES · BALANCED FRAMING

OPTIONAL ECOLOGICAL PROCESS DEPTH
- CAUTION: Effective air-pollution policy depends on accurate source apportionment (scientifically
- CAUTION: Common analytical error to avoid: treating crop-residue burning as the dominant,
- NCAP set city-specific particulate-matter (PM2.5/PM10) reduction targets for identified
- CAUTION: Implementation critique: progress tracking under NCAP has faced challenges

SCALE, PARAMETER OR STATUS LIMIT
- CAUTION: Balanced framing: NCAP represents a scientifically sound target-setting framework;
- CAUTION: Real-time AQI readings depend on monitoring-station density and placement; sparse
- CAUTION: NAAQS permissible limits are themselves periodically reviewed against evolving health
- CAUTION: Attributing health outcomes (mortality, morbidity) precisely to a specific pollution

COMPARATIVE RESTORATION NUANCE
- The Commission for Air Quality Management in NCR and Adjoining Areas Act, 2021
- GRAP is a reactive, severity-triggered emergency-response mechanism; NCAP is a
- Source apportionment studies show the relative contribution of crop-residue burning to
- CAQM was established partly because prior coordination on Delhi-NCR air pollution had

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels